INCH-POUND

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MILITARY SPECIFICATION

BORING MACHINES, JIG, VERTICAL, SINGLE COLUMN AND DOUBLE COLUMN BRIDGE TYPES, MANUAL AND POWER FEED

1. SCOPE

1.1 Scope. This specification covers floor mounted, vertical spindle, jig boring machines with manually operated feed devices and manually controlled power axis drives with equipment and accessories specified herein.

1.2 Classification. The machines shall be of the following types, styles, and sizes. The type, style, and size to be furnished shall be as specified (see 3.5 and 6.2.1).

Type I - Manually operated feed devices.

Style A - Single column.

Size	Table cross travel, inches	Table longitudinal travel, inches
1118	11	18

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3411

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Type II - Manually controlled power axis drives.

Style A - Single column.

Size	Table cross travel, inches	Table longitudinal travel, inches
808	8	8
1016	10	16
1118	11	18

Style B - Double column.

Size	Table cross travel, inches	Table longitudinal travel, inches
1825	18	25
2436	24	36
2448	24	48
2452	24	48 (Extended table size)

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. The issue of these documents shall be the latest revision of the documents on the date of the solicitation and may include title changes.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

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MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents and publications. The following other Government document forms a part of this specification to the extent specified herein. The issue shall be the issue in effect on the date of the solicitation and may include title changes.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of these documents shall be the latest revision of the document on the date of the solicitation and may include title changes.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI B11.8 - Safety Requirements for the Construction, Care, and Use of Drilling, Milling, and Boring Machines.

ANSI/ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

ANSI/NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of drilling, boring, reaming, and light milling on jigs, dies, fixtures, and precision production parts in accordance with the requirements herein. Type I machines require manual positioning. Type II machines position by power leadscrew to coordinates determined by the operator. Style A machines are of single column construction; style B machines are of double column and bridge construction. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

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3.2.1 Design and construction measurement systems. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.2 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in US, SI, or dual (US and SI) units as specified by the procuring activity (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.001 inch or degrees and minutes in increments not larger than one-half minute. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2.2 Digital readout (DRO) system. When specified (see 6.2.1), a DRO system shall be installed on the X (longitudinal) and Y (cross) axes of travel. The DRO system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to assure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.00005 inch. The system accuracy shall be equal to the system resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0001 inch, measured over the full travel on each axis. Special requirements for the DRO system components shall be as specified (see 6.2.1). The DRO shall indicate axis location in both the US and SI units. The DRO system shall be provided with electronic and manual calibration to the machine on which it is installed to eliminate the effects of distortion error. An input power source fluctuation of $\pm$ 10 percent shall not adversely affect the DRO system operation.

3.2.2.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on

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any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines in any multiple machine order.

3.2.2.2.2 Electronic display console. The electronic display console for the DRO system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keyboard entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's work position. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be armored and shall be installed so as not to interfere with the normal operations of the machine. The cable shall be securely fastened to the structure of the machine in accordance with the requirements of the contract.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator's work station(s). Spindle speed and direction of rotation, feed rates, axis motion drive, coolant, and start and stop switches, as applicable, shall be operated from the operator's normal work station. Control devices shall be functionally grouped and clearly identified. An emergency control to stop all machine operations shall be provided.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for the point of operation and all other parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of 29 CFR 1910 and ANSI B11.8. If a conflict arises between 29 CFR 1910 and ANSI B11.8, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

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3.2.8 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Means shall be provided to indicate a low oil condition. All oil holes, grease fittings, and filler caps shall be accessible. When specified (see 6.2.1), an automatic lubrication system shall be provided.

3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts (except for hand-fitted parts such as ways and boxed bearings) shall be interchangeable without requiring modification. Replacement parts are those which are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include, but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall be interchangeable with domestically supplied components or be logistically supportable and readily available from a domestic source.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as weldings, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

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3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. Except for leadscrews, when applicable, all threaded parts, either US or SI, used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears, shafts, and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the US system or ANSI/ISO 54 for the SI system. The gears, shafts, and pinions shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears, shafts, and pinions in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Machine ways. All machine ways shall be protected to exclude chips, coolant, and other hazards that may cause damage or result in inaccurate functioning of any members. Integrally cast ways shall be hardened, ground, and hand fitted to meet the accuracies required in table II. Replaceable or insert type ways shall be hardened to Rockwell C-55 and precision ground to meet the accuracies required in table II. A metered automatic lubrication system shall be provided for all ways. Each lubricant reservoir shall have at least a 24 hour capacity and means for determining fluid level.

3.3.9 Slides. All sliding members shall have means of adjustment, as necessary, to compensate for the wear of mating components and for wear of the mating surfaces and to meet and maintain the accuracy and alignment tolerances given in table II.

3.3.10 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist basically of the following components. Any additional components necessary for the operation of the machine shall be furnished in accordance with the manufacturer's standard practice.

3.4.1 Base. The base shall be a casting of iron or iron alloy, ribbed and braced to minimize distortion, vibration, and deflection. The base shall form a foundation for the entire machine. The base shall have proportioned guideways to fully support the saddle on style A machines and the worktable on style B machines throughout their full range of travel. All anchor bolts and leveling pads required to level and secure the machine to the floor shall be provided. Hydraulic, coolant, and electrical systems shall be separate external units. When applicable, access doors or removable plates shall be provided to service components housed within the base.

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3.4.2 Column assembly. Each column shall be a one-piece casting of iron or iron alloy of boxlike construction, internally ribbed and braced to minimize distortion and deflection under normal tool loads. The column shall be either integrally cast with the base or separately constructed, machined, and securely attached to the base. The column on style A machines shall accommodate the main spindle housing assembly and its associated components. Guideways shall support and guide the spindle housing assembly through its full range of vertical adjustment (W axis). Means shall be provided to lock the spindle assembly in any location within its vertical range. Means shall be provided to return the spindle housing assembly to the same location after assembly movement. Stops or limit switches located at the extremes of spindle assembly movement shall prevent overtravel of the assembly. Style B machines shall meet the same basic requirements as style A machines except that they shall have a double column, bridge, and sliding cross carriage configuration accommodating the main spindle assembly.

3.4.3 Bridge assembly. Type II, style B machines shall be constructed with a cast iron bridge-like member. It shall be internally ribbed and braced to minimize distortion and deflection experienced under normal operating loads. The bridge shall be fixed, supported on two vertical box-type columns located on either side of the base casting, simulating a bridge over the base and worktable. On machines without a cross rail, horizontal guideways shall be provided on the bridge to accept a moveable cross carriage assembly and to allow the cross carriage to move in the transverse (Y axis) direction. On machines with a cross rail, vertical guideways shall be provided in the columns supporting the cross rail to allow the cross rail to traverse in the vertical (W axis) direction. Stops shall be provided to prevent overtravel. The cross rail shall be equipped with fully automatic, distortion-free clamping. Horizontal guideways shall be provided on the cross rail to accept a cross carriage and to allow the cross carriage to move in the transverse (Y axis) direction. The cross rail shall be balanced to ensure parallel movement of the spindle headstock in relation to the worktable.

3.4.4 Cross carriage. Type II, style B machines shall be provided with a cross carriage. The cross carriage shall be made of iron or iron alloy, precision fitted to the bridge or cross rail, traveling in the transverse (Y axis) direction. The cross carriage on machines without a cross rail shall be provided with vertical guideways to accommodate the spindle housing assembly and its associated components and to allow the assembly to travel vertically (W axis). On machines with a cross rail, the cross carriage and spindle housing assembly shall travel in the transverse (Y axis) direction on horizontal guideways on the cross rail and vertically (W axis) on vertical guideways in the columns. An automatic, non-influencing clamping system shall be provided to secure cross carriage location after positioning. When specified (see 6.2.1), the cross carriage (Y axis) shall be provided with power and rapid traverse feed.

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3.4.5 Machining head assembly. The machining head assembly shall include, but not be limited to, the quill (as required), spindle with corresponding feed and speed mechanisms, vertical slide handwheel, spindle housing clamp, adjustable depth stop, and related components necessary for drilling, tapping, milling, and boring. The assembly shall be made of nickel iron alloy with a low coefficient of thermal expansion. Means shall be provided on style A machines to thermally stabilize the machining head to minimize distortion. The machining head shall hold the spindle in axial alignment perpendicular to the work mounting surface of the worktable and shall prevent distortion of the head due to torsional forces of the drives. The assembly shall be mounted on the column or cross carriage guideways, as applicable, and shall travel vertically (W axis). The assembly shall be balanced and counterbalanced. A repeatable clamping mechanism shall be provided to lock the assembly to the column at any position within its vertical travel. A handwheel shall be provided for manual adjustment of the boring head assembly vertical position (W axis). Power vertical head rapid adjustment (W axis) shall be provided when specified (see 6.2.1). Stops located at the extremes of vertical travel shall be provided to prevent overtravel of the head assembly. Means shall be provided to indicate vertical assembly location.

3.4.5.1 Machining head spindle. The spindle shall be of alloy steel, hardened to at least Rockwell C-55 and ground to a finish not rougher than 32 microinch. The spindle shall be supported and held in axial and radial alignment by two or more sets of preloaded ball or roller bearings. The spindle shall be mounted in the head with its bearings arranged to compensate for thermal expansion. The spindle shall have positive means for accepting, retaining, and releasing tools and tool holders inserted by the operator. Unless otherwise specified (see 6.2.1), nose, bores, and other spindle characteristics shall be the manufacturer's standard design. The spindle shall rotate clockwise and counterclockwise under the full range of spindle speeds specified in table I. Provisions shall be made to adjust the spindle speed in either continuous adjustment or in steps of not greater than 10 percent. The machine overload protection shall automatically stop spindle rotation should the machine stall from overload. A depth-measuring device shall be provided to indicate spindle vertical position (Z axis). A handwheel shall be provided for spindle fine feed.

3.4.5.1.1 Automatic spindle cycle. When specified (see 6.2.1), an automatic spindle cycle shall be provided. This cycle shall include starting spindle rotation while rapid advancing to the workpiece (60 IPM minimum), engaging feed, stopping feed at a depth preset by the operator, and spindle retraction to the starting position when the cycle is complete. An adjustable, automatic depth trip stop shall be provided for use with this automatic spindle cycle for presetting the machining depth. The automatic spindle cycle shall meet the depth (Z axis) positioning accuracy and repeatability requirements given in Table III.

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3.4.6 Saddle. Style A machines shall be provided with a saddle. The saddle shall be a casting of iron or iron alloy, constructed to withstand all loads which may be imposed during normal operation of the machine. The saddle shall be precision fitted to the guideways of the base and shall travel in the transverse (Y axis) direction. Longitudinal guideways shall be provided on the saddle top to accept a sliding worktable. The saddle shall operate on a leadscrew or precision ballscrew with a recirculating ball nut. A non- influencing clamping system shall be provided to prevent movement of the saddle after positioning and during machining operations. Stops shall be provided at both ends of saddle travel to prevent overtravel. Saddle functions on type I machines shall be controlled manually by the operator. When specified (see 6.2.1), operator controlled saddle power and rapid traverse feed on type II machines shall be provided.

3.4.7 Worktable. The worktable shall be a casting of iron or iron alloy, proportioned to support and withstand maximum capacity workpiece weights and tool thrust loads. On style A machines, the worktable shall be mounted on the saddle; on style B machines, the worktable shall be precision fitted to the base ways. The worktable shall traverse longitudinally (X axis). A non-influencing clamping system shall be provided to prevent movement of the worktable after positioning and during machining operations. Means shall be provided at each end of travel to prevent worktable overtravel. The surface of the worktable shall be hand or machine finished for a work mounting surface. The worktable shall have a coolant trough or similar means for draining spent coolant into its reservoir. Unless otherwise specified (see 6.2.1), the number of T-slots shall be manufacturer's standard. The size and configuration of T-slots shall meet ANSI B5.1. The table shall be provided with means of securing workpieces in place to permit precision machining without drifting. Type I worktable movement shall be controlled manually by the operator. Type II machine worktables shall be provided with power and rapid traverse feeds controlled manually by the operator. When specified (see 6.2.1), the worktable shall be furnished with a subplate with characteristics as specified.

3.4.8 Positioning system. When specified (see 6.2.1), type II machines shall be provided with a precision 2-axes positioning and measuring system of the manufacturer's standard type. The system shall provide accurate 2-axes feedrate and simultaneous positioning control of the X and Y axes. The system shall allow positioning of the X and Y axes manually, by jogging (the operator predetermines axes movement), by teach-in (positions recorded by the system during a manual run), and automatically (axes movements stored in the system by the operator and then run automatically). The system shall also measure and check set axes positions. The system display console shall meet the requirements of 3.2.2.2.2 and shall display actual and preset X and Y axes location. The system shall meet the positioning accuracy requirements given in table III.

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3.4.9 Coolant system. Unless otherwise specified (see 6.2.1), the machine shall be provided with a flood-type coolant system to deliver a flow of coolant to the tool. The coolant system shall include a sump or reservoir, a power driven pump, splash guards, drain troughs, and all necessary piping. The reservoir shall have means to prevent chips from entering the pump. The coolant pump motor shall be independent of the main drive motor of the machine. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for regulating the flow of flood coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning.

3.4.10 Hydraulic system. The hydraulic system, when supplied, shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.11 Electrical system. The electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit having a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts.

3.4.11.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), alternating-current (AC) motors shall be designed to operate on 60-Hz and each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of ANSI/NEMA MG-1. Each motor shall have an identification plate to identify manufacturer, model identification, serial number, voltage, amperage, horsepower, phase, and frequency.

3.4.11.2 Worklight. Unless otherwise specified (see 6.2.1), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.11.3 Electrical load indicator. When specified (see 6.2.1), the machine shall have a meter to indicate the electrical load on the spindle drive motor. The meter shall be current actuated. The dial shall indicate either amperes or percent of rated full-load current, and shall be clearly marked to indicate the rated reading for continuous operation of the machine. The accuracy of the meter shall be within 2 percent of full-rated load. The meter shall be clearly readable from the operator's position.

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3.4.11.4 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4.12 Drive mechanisms. Means shall be provided to position each moving member in all axes of motion. All leadscrews shall be hardened, ground, and corrective lapped and all ballscrews shall be hardened and precision ground. Leadscrews and ballscrews shall be of a size, class, and precision suitable for the application. Leadscrews or ballscrews and nuts shall not exert excessive loads or cause premature deterioration of the feed drive components. A metered automatic lubrication system or oil bath shall be provided for all axis screws. Each lubricant reservoir shall have at least a 24 hour capacity and means for determining fluid level. The slides and screws shall be protected from foreign elements by covers.

3.4.12.1 Manual movement. Type I machines shall position along each coordinate axis by manual means, using handwheels and precision ballscrews or leadscrews. Movement shall be initiated and controlled by the operator.

3.4.12.2 Power movement. Type II machine components shall move along each coordinate axis by either manual means using handwheels or by power leadscrew controlled manually by the operator. For movement, the system shall require the operator, by single pushbutton, lever, or selector switch, to locate the spindle by power to a chosen position on the workpiece. The system shall be permanently built into the machine in a manner to prevent contamination which would affect the accuracy of positioning. This system, through operator control of each axis, shall enable the drives to rapid traverse and decelerate to the actual desired positions without overshoot.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than specified in table I for the type, style, and size specified.

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II when tested using the procedures given in 4.8.1.

3.7 Performance. The machine shall be capable of boring, drilling, reaming, and light milling when tested using the procedures given in 4.8.2 through 4.8.3.2.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), all standard equipment normally furnished by the manufacturer shall be furnished with each machine.

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3.9 Optional equipment. The following equipment and accessories are commercially available and are considered optional. When required (see 6.2.1), optional equipment shall be furnished as specified and fully described.

a. Collet holder with complete set of spring collets for fractional sizes, for end mills.

b. Drill chuck, keyless, of the capacity specified

c. Boring head, of the type and taper specified.

d. Facing and boring head, of the type and taper specified.

e. Indicator complete with holder of the capacity specified.

f. Dial indicators for depth setting, feeds, boring head, and facing head.

g. Precision rotary table.

h. Parallel set-up blocks.

i. Precision vise.

j. Microscope.

k. Centering and aligning tools.

l. Dust extractor.

m. Storage cabinet.

n. Other optional equipment as specified (see 6.2.1).

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2.1). All repair parts shall be assigned a part number identified in a repair parts manual.

3.11 Marking on instruments, control panels, and plates. All words on instruments, control panels, and plates shall be in the English language. Characters or international symbols shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion-resistant.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

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Points of lubricant application
Servicing interval
Type of lubricant
Viscosity
Military or Federal specification number for each lubricant
Size and type of each lubricant filter

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the machine shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.6).

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in an indoor facility with an ambient temperature of 50°F to 100°F and 10 percent to 95 percent relative humidity.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.7, all tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on the machine to ensure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The machine shall be examined to determine compliance with all requirements of this specification including compliance with all applicable documents cited in Section 2.

4.8 Tests. All instruments, materials, and tools required to perform and evaluate the following tests shall be furnished by the manufacturer. The measuring instruments used shall have evidence of calibration traceable to the National Bureau of Standards (NBS). At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the procuring agency in lieu of or in addition to the following tests (see 6.2.1).

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4.8.1 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The methods used in performing the tests shall be at the option of the supplier provided they accurately determine the tolerances of the machine. The alignment tolerance test shall be performed prior to and subsequent to the performance tests. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.2 Operational test. The machine shall be operated at no load for not less than 2 hours. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Performance tests. The plate used for the performance test shall be of a material chosen by the manufacturer, not less than 1 inch thick. Both faces may be surface ground prior to the test. Drill and bore three holes, 3/4-inch diameter by 3/4-inch deep, in a right angle pattern in the metal plate. The dimensions between the bored holes shall be spaced to use the maximum normal working range of the machine, allowing for set-up devices. Counterbore the hole 1 inch by 1/8-inch depth. The workpiece shall be removed from the machine for measuring. Counterbores shall be concentric with bored holes within 0.0001 inch TIR. All machined hole diameters shall be alike within ±0.0001 inch TIR. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.8.3.1 Positioning system test. When type II machines are provided with a positioning system (see 3.4.8), the system shall be tested for positioning accuracy. Using a dial indicator and the positioning system, relocate the bored holes, two in-line on each axis. The positioning system should be cycled a minimum of three times with the positioning accuracy recorded for each cycle. The machine shall relocate each hole to within ±0.00032 inch.

4.8.3.2 Automatic spindle cycle test. When the machine is provided with an automatic spindle cycle (see 3.4.5.1.1), the cycle shall be tested for depth positioning accuracy and repeatability. The cycle shall be run a minimum of three times. The cycle shall position the spindle to within the requirements given in table III.

4.8.4 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.10 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

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6. NOTES

6.1 Intended use. The jig boring machines covered by this specification are intended for drilling, boring, reaming, and light milling on jigs, dies, fixtures, and precision parts in any tool room or production shop to locate, drill, bore, tap, or measure parts to precision tolerances.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, style, and size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. Specify system of units for measuring and indicating devices (see 3.2.2).

f. Digital readout system, if required; specify axes of travel and any special requirements (see 3.2.2.2).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Automatic lubrication system, if required (see 3.2.9).

i. Painting, if different (see 3.3.5).

j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.10).

k. Power cross carriage on type II, style B machines, if required (see 3.4.4).

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l. Power head adjustment (W axis), if required (see 3.4.5).

m. Nose, bores, and other spindle characteristics, if different (see 3.4.5.1).

n. Automatic spindle cycle, if required (see 3.4.5.1.1).

o. Power saddle feed (Y axis) on style A machines, if required (see 3.4.6).

p. T-slots, if different (see 3.4.7).

q. Specify worktable subplate characteristics, when required (see 3.4.7).

r. Positioning system, if required (see 3.4.8).

s. Coolant, if not required (see 3.4.9).

t. Electrical system, if different (see 3.4.11).

u. Motors, if different (see 3.4.11.1).

v. Work light, if not required (see 3.4.11.2).

w. Electrical load meter, if required (see 3.4.11.3).

x. Hour meter, if required (see 3.4.11.4).

y. Size and capacity, if different (see 3.5).

z. Standard equipment, if different (see 3.8).

aa. Specify and fully describe optional equipment, if required (see 3.9).

bb. Specify applicable repair parts, if required (see 3.10).

cc. Lubrication plate, if different (see 3.11.1).

dd. Nameplate, if different (see 3.11.2).

ee. Inspection conditions, if different (see 4.3).

ff. First article inspection, if different (see 4.4).

gg. Quality conformance inspection, if different (see 4.5).

hh. Acceptance test, if different (see 4.6).

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ii. Tests, if different (see 4.8).

jj. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts, lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross-reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

MIL-B-9925C	MIL-B-9925D
Size 914	Deleted
Size 1118	Type I, Style A, Size 1118
Size 1216	Deleted
Size 1630	Deleted
Size 1828	Deleted
Size 2036	Deleted
Size 2236	Deleted
Size 2446	Deleted
Size 2848	Deleted
Size 3660	Deleted
New	All Type II machines

6.5 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Inspection facilities. The contracting officer should be notified by each contractor in his response to the solicitation of the location where inspection tests are to be performed. It is recommended the contract requires all inspections be performed within the limits of the continental United States.

6.7 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.8 Warranty. Warranty requirements shall be as specified by the procuring activity in the contract.

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Table I. Sizes and capacities.

Characteristic	Requirements			
	Type I	Type II		
	Style A	Style A		
	Size 1118	Size 808	Size 1016	Size 1118
Capacity:				
Table working surface, width x length, inches	11 x 24	9-1/2 x 14-1/4	15 x 23-5/8	11 x 24
Table longitudinal travel (X axis), inches	18	8	16	18
Table to spindle nose, inches	3-7/8 to 20-13/16	0 to 13-5/8	0 to 21-5/8	3-7/8 to 20-13/16
Saddle cross travel (Y axis), inches	11	8	10	11
Head travel (W axis) on column, inches	12	9-1/2	18	12
Spindle quill travel (Z axis), inches	5	2-3/4	5-1/2	5
Spindle center to column, inches	10	10-1/4	12-3/4	10
Maximum load, pounds	500	175	650	500
Maximum drill diameter	1	3/8	1-1/4	1
Maximum bore diameter	5	2	8-5/8	5
Speeds and feeds:				
Spindle speeds, number	← Infinitely variable →			
Spindle speed range, RPM	60 to 2250	150 to 3000	30 to 4000	60 to 2250
Spindle feeds, number	3	1	4	3
Spindle feed range, inches per revolution	0.00075 to 0.003	0.0001 to 0.0006	0.0008 to 0.009	0.00075 to 0.003
Head rapid traverse, inches per minute	—	10	35	—
** Worktable and saddle rapid traverse, inches per minute	—	10	59	16
Motors, HP:				
Spindle drive	1 or 2	5	5	1 or 2

**When applicable.

NOTE 1: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in the above table. When the largest size and capacity machine in the above table is ordered, the size and capacity of the machine shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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Table I. Sizes and capacities - continued.

Characteristic	Requirements Type II Style B			
	Size 1825	Size 2436	Size 2448	Size 2452
Capacity:				
Table working surface, width x length, inches	30 x 24-3/4	40 x 34	24 x 48	52 x 34
Table longitudinal travel (X axis), inches	25-1/2	35-1/2	48	48
Table to spindle nose, inches	26-1/4	33-1/2	28	33-1/2
Head cross travel (Y axis), inches	18	23-1/2	24	23-1/2
Head travel (W axis) on column, inches	19-1/2	25	13	25
Spindle quill travel (Z axis), inches	6-1/4	7-3/4	8	7-3/4
Clearance between columns, inches	29-1/2	37-1/2	32	37-1/2
Maximum load, pounds	1100	1760	2000	1760
Speeds and feeds:				
Spindle speeds, number	← Infinitely variable →			
Spindle speed range, RPM	30 to 3000	30 to 3000	50 to 500	30 to 3000
Spindle feed range, inches per revolution	0.0004 to 0.0153	0.0004 to 0.0153	0.0002 to 0.020	0.0004 to 0.0153
Head rapid traverse, inches per minute	240	240	16	240
** Worktable and saddle rapid traverse, inches per minute	240	240	16	240
Motors, HP:				
Spindle drive	5	5	5	5

**When applicable.

NOTE 1: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in the above table. When the largest size and capacity machine in the abo table is ordered, the size and capacity of the machine shall not exceed the stated requirements by more than 10 perce When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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Table II. Alignment and accuracy tolerance requirements.

Requirement	Permisible tolerance
Head and Spindle Assembly:	
Spindle centerline parallel to quill centerline	0.00015" TIR
Spindle axial slip (in 360 degrees rotation)	0.0003"
Concentricity of spindle nose internal bore, 3/8" from nose	0.00005" TIR
Spindle bore runout (in 360 degrees rotation)	
- 1/4" from nose	0.0001" TIR
- 12" from nose	0.0005" TIR
Spindle axis square with worktable surface (in 180 degree sweep at 6")	0.0003"
Quill motion square to worktable surface (with the boring head locked)	
- quill movement (Z) alone	0.00015"/10"
- head movement (W) alone	0.00015"/10"
Quill dynamic load deflection (at 2000 RPM)	0.0003"
Head cross travel (Y axis) perpendicular to cross rail vertical travel (W axis)	0.0003"/ft
Transverse rise or fall of head	0.0006"/ft
Worktable and Cross Rail Assembly:	
Worktable trueness of travel	0.0003" overall
Worktable longitudinal travel (X axis) perpindicular to cross travel (Y axis)	0.0001"/ft 0.0003" overall
Worktable flat in all directions	0.0001"/ft 0.0002" overall
T-slots parallel to each other	0.001"/ft 0.0015" overall
Worktable trueness of travel	
- Sizes 808 through 1118	0.0002" overall
- Size 1825	0.0003" overall
- Sizes 2432 through 2452	0.0005" overall
Cross rail motion perpendicular to worktable surface (in 25")	
- left to right	0.00032"
- front to rear	0.00048"
Longitudinal rise or fall of worktable	0.0002"/ft

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Table III. Positioning accuracy and repeatability.

Requirement	Permissible tolerance
** Worktable (X and Y axes) positioning accuracy (each direction):	
- Sizes 808 through 1118	± 0.00024" overall
- Sizes 1825 and larger	± 0.00032" overall
** Worktable positioning repeatability (X and Y axes) (each direction)	± 0.00004"
++ Depth positioning (Z axis) accuracy	± 0.0005"
++ Depth positioning (Z axis) repeatability	± 0.0002"

**Applicable only to type II machines with the optional positioning system (3.4.8).

++Applicable only on machines with the optional automatic spindle cycle (see 3.4.5.1.1).

6.9 Subject term (keyword) listing.

Boring machine
Bridge type
Column, double, single
Cross rail, fixed, traveling
Digital readout
Jig
Manual
Power
Vertical spindle

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA - IP

Project (3411-0062)

Review activities:
Army - AR
Navy - OS
Air Force - 84

User activities:
Army - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-9925D	Boring Machines, Jig, Vertical, Single Column and Double Column Bridge Types, Manual and Power Feed

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: –
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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